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Q7)  $f: A \rightarrow B$

$$C \subseteq A$$

$$f|_C = f \cap (C \times B)$$

(a)

$f|_C$  is a function:  $C \rightarrow B$

$$\forall c \in C \exists ! b \in B (c, b) \in f|_C$$

let  $c \in C$  arbitrary.

$$\exists ! b \in B (c, b) \in f \cap (C \times B)$$

Existence: Note that  $c \in A$  as well

let  $b = f(c)$  so  $b \in B$ . so  $(c, b) \in C \times B$  & as  $C \subseteq A$  so

$$c \in A \text{ so } (c, b) \in f. \text{ so } (c, b) \in f \cap (C \times B)$$

Uniqueness:

$$\forall b_1 \in B [(c, b_1) \in f|_C \rightarrow b_1 = b]$$

supp  $b_1 \in B$

$$\& (c, b_1) \in f|_C$$

$$b_1 = b$$

$$\text{so } (c, b_1) \in f \cap (c, b_1) \in C \times B$$

but we know  $(c, b) \in f$  &  $f$  is a function so it must be that  $b = b_1$ .

To derive formula  $f|_C(c) = (f|_C \times c)$

~~(f|\_C \times c)~~

supp  $c \in C$

$\rightarrow$  supp  $f(c)$

$$\rightarrow (f|_C \times c)$$

$$\text{let } b = f(c) \in B$$

$$f \cap (C \times B)(c)$$

$$\text{so } (c, b) \in f$$

$$(c, b) \in f \cap (C \times B)$$

clearly  $c \in C$  &  $b \in B$  so  $(c, b) \in C \times B$  & we already know

that  $(c, b) \in f$ .

since we already proved it's a function

$$\leftarrow \text{supp } (c, b) \in f|_C \text{ clearly } (c, b) \in f \text{ as } f|_C$$

$$= f \cap (C \times B).$$



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(b)  $g: C \rightarrow B$   $g = f \circ c \iff g \leq f$   
 $\rightarrow \text{supp } g = f \circ c$   $\rightarrow g \leq f$   
 $\text{Supp } (c, b) \in g$  in arbitrary  $(c, b) \in f$   
 $\wedge g = f \circ c$  so  $(c, b) \in f \circ c$  so  $(c, b) \in f \cap (C \times B)$   
 so  $(c, b) \in f$ .  
 $\leftarrow \text{supp } g \leq f$   $g = f \circ c$   
 Any arbitrary  $(c, b) \in g$  iff  $(c, b) \in (C \times B) \cap (c, b) \in f \circ c$   
 iff  $(c, b) \in f \cap (C \times B)$  iff  $(c, b) \in f$

(c)  $g = \{(x, y) \in \mathbb{Z} \times \mathbb{R} \mid y = 2x + 3\}$   
 $h = \{(x, y) \in \mathbb{R} \times \mathbb{R} \mid y = 2x + 3\}$   
 $h(x)$

$g = h \circ 2$   
 $\rightarrow \text{supp } (x, y) \in g$   $(x, y) \in h \circ 2$   
 so  $(x, y) \in \mathbb{Z} \times \mathbb{R} \wedge y = 2x + 3$   $(x, y) \in h \cap (\mathbb{Z} \times \mathbb{R})$   
 so so that  $(x, y) \in h$   
 as  $\mathbb{Z} \subseteq \mathbb{R}$  so  $x \in \mathbb{R}$  as well so  $(x, y) \in \mathbb{R} \times \mathbb{R}$  as well  
 $\in y = 2x + 3$  as we already know.  
 $\leftarrow \text{supp } (x, y) \in h \circ 2$   $(x, y) \in g$   
 iff  $(x, y) \in \mathbb{R} \times \mathbb{R} \cap (x, y) \in \mathbb{Z} \times \mathbb{R} \wedge y = 2x + 3$   
 clearly goal follows.

$y = 2(x)$

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Q8.  $f: A \rightarrow B$   $\exists A' \subseteq A (g: A' \rightarrow B)$   
 $g \leq f$   $g \subseteq f$   
 let  $A' = \{a \mid (a, b) \in f\} = \text{Domain of } f$   
 let  $a \in A'$  in arbitrary as  $(a, b) \in f \wedge g \leq f$   
 so  $(a, b) \in g$  so  $a \in A'$  so  $A' \subseteq A$ .  
 To see that  $g$  is a function  $\forall a \exists! b \in B (a, b) \in g$   
 let  $a \in A'$   $A' \subseteq A$  so  $a \in A$  as well  
 let  $b = f(a) \in B$  so  $(a, b) \in f$   
 then by  $\circ$  we defined there exist some  $b \in B$  with  $(a, b) \in f$

Uniqueness  
 $\text{supp } (a, b_1) \in g \wedge (a, b_2) \in g$   $b_1 = b_2$   
 so  $a \in A'$   $b_1, b_2 \in B$   $\wedge g \leq f$   
 as  $A' \subseteq A$  so  $a \in A$   
 so  $(a, b_1) \wedge (a, b_2) \in f$  but  $f$  is a function so  $b_1 = b_2$

(d)  $g: A \rightarrow B$   
 $B \neq \emptyset$   $\exists g: A' \rightarrow B (A \leq g)$   
 $A \subseteq A'$   
 By analysing an example we can see that we don't want to repeat the domain.  $\mathbb{Z} \subseteq \mathbb{R} \neq \emptyset$  so let  $b \in B$  - but this is only clear that we will map elements of  $A'$  so domain won't be repeated. Also  $g$  should contain  $f$ . So after analysing an example  $g = f \cup ((A' \setminus A) \times \{b\})$  looks like the right choice



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we will prove first that  $g = A' \rightarrow B$   
 existence: supp  $a \in A'$ . Then there is  $b_0 \in B$  such  
 that by defn of  $g$ ,  $(a, b_0) \in g$ .

Uniqueness:

supp  $(a, b_1) \in g \wedge (a, b_2) \in g$ .  $b_1 = b_2$

$\because$  As  $g = f \cup \{(A' \setminus A) \times \{b_0\}\}$ . We consider 2 cases

case 1) supp  $(a, b_1) \notin f$ .

case 1.1)  $(a, b_1) \in f$ . The  $f$  is  $f: A \rightarrow B$  so  $b_1 \in B$ .

case 1.2)  $(a, b_2) \in \{(A' \setminus A) \times \{b_0\}\}$  then  
 $a \in A' \setminus A$  which is contradiction over  $a \in A$   
 as  $(a, b_1) \in f$ . so this case is not possible.

case 2)  $(a, b_1) \in \{(A' \setminus A) \times \{b_0\}\}$ . then  $b_1 = b_0$

if  $(a, b_2) \in f$ . then  $a \in A$  which is contradiction  
 over  $a \notin A$ . so this is not possible.

so  $(a, b_2) \in \{(A' \setminus A) \times \{b_0\}\}$  so  $b_2 = b_0$

so  $b_1 = b_2 = b_0$ .

lemma proved.

Finally to see that  $f \subseteq g$ .

supp  $(a, b) \in f$ . then  $\rightarrow (a, b) \in g$ .  
 as  $g = f \cup \{(A' \setminus A) \times \{b_0\}\}$  so  $(a, b) \in g$ .

(R10)  $f: A \rightarrow B, g: A \rightarrow B, f \neq g$ .

$f \cup g$  is not a function  
 $(f \cup g) \setminus (f \cap g)$  is not a function

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by translating logical symbols, we can see that as  $f \neq g$   
 we can choose some  $(a, b) \in f \setminus g$  such that  $(a, b) \notin g$   
 $\forall (a, b) \in g \wedge (a, b) \notin f$ .

Case 1)  $(a, b) \in f \setminus g$ .

We will use more that  $f = g$  iff  $\forall a \in A (f(a) = g(a))$   
 as  $f \neq g$ . so we can choose

some  $a_0 \in A (f(a_0) \neq g(a_0))$

give  $f: A \rightarrow B$  so  $(a_0, f(a_0)) \in f$ . as  $g$  is a function.

so similarly  $(a_0, g(a_0)) \in g$ . so  $(a_0, f(a_0)) \in f \setminus g$

then  $f(a_0) \neq g(a_0)$  so  $f \neq g$  is not a function.  
 similar argument

(R11) we have already seen that it is equivalent in (R1.4) &  
 we saw in (R1.5) it's a function.

Uniqueness:

supp  $f$  &  $g$  are 2 functions on  $A$  that are also equivalent  
 relations  $\rightarrow f = g$

so  $f: A \rightarrow A, g: A \rightarrow A$

let  $a \in A$ .

By reflexivity of  $f$  &  $g$ ,  $(a, a) \in f$  &  $(a, a) \in g$   
 so  $f(a) = a = g(a)$  & hence proved

(R12)



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$$f: A \rightarrow C$$

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$$Q12) (b) \rightarrow \text{supp } f \cup g : A \cup B \rightarrow C$$

$$\begin{aligned} & \text{supp } (x, y) \in f \cup g \iff (x, y) \in f \vee (x, y) \in g \\ & \text{supp } (x, y) \in f \iff (x, y) \in f \cap (A \cup B) \times C \\ & \text{supp } (x, y) \in g \iff (x, y) \in g \cap (A \cup B) \times C \\ & \text{As } x \in A \cap B \text{ so } x \in B \\ & \text{As } g: A \rightarrow C \end{aligned}$$

$$\begin{aligned} & \text{so } \exists y_1 \in C \text{ s.t. } (x, y_1) \in g \\ & \text{If we can prove } y_1 = y \text{ then we are done} \\ & \text{As } x \in A \cap B \text{ so } x \in A \\ & \text{so } \exists y_2 \in C \text{ s.t. } (x, y_2) \in f \\ & \text{Case 1) } (x, y_1) \in g \text{ then } y = y_2 \text{ (if } f \text{ is } f \text{ then)} \\ & \text{Case 2) } (x, y_2) \in f \text{ then } y = y_2 \\ & \text{so } y = y_2 \\ & \text{so we have } (x, y) \in f \cup g \text{ for } A \cap B \rightarrow C \\ & \text{so we have } f \cup g \text{ is a function} \end{aligned}$$

$$\begin{aligned} & \text{supp } f \cup g : A \cup B \rightarrow C \\ & \text{As } f: A \rightarrow C \text{ and } g: B \rightarrow C \\ & \text{so } f \cup g : A \cup B \rightarrow C \\ & \text{As } (a, c_1) \in f \text{ so } (a, c_1) \in f \cup g \text{ similarly } (a, c_2) \in f \cup g \\ & \text{but } f \cup g : A \cup B \rightarrow C \text{ so } c_1 = c_2 \end{aligned}$$

Note: The problem was that I was considering cases that  $a \in A$  so  $a \in A \cup B$ . The truth is 2 cases,  $a \in A$  or  $a \in B$ . But we should also try  $f \cup g$  as well. As  $A \cap B$ , so using only it is not as powerful as using  $f \cup g$ .

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$$\begin{aligned} & \leftarrow \text{supp } f \cup g : A \cup B \rightarrow C \\ & \text{supp } f \cup g : A \cup B \rightarrow C \\ & \text{Case 1) } a \in A \text{ as } f: A \rightarrow C \text{ let } f(a) = c_1 \in C \text{ so } (a, c_1) \in f \cup g \\ & \text{Case 2) similar} \\ & \text{Uniqueness:} \\ & \text{supp } (a, c_1) \in f \cup g \text{ and } (a, c_2) \in f \cup g \text{ there are 4 cases.} \\ & \text{The cases when } (a, c_1) \in f \text{ and } (a, c_2) \in f \text{ so } c_1 = c_2 \\ & \text{Case 3) } (a, c_1) \in f \text{ and } (a, c_2) \in g \\ & \text{so } a \in A \text{ and } a \in B \text{ so } a \in A \cap B \text{ as } f \text{ is } f \text{ then} \\ & \text{so } (a, c_1) \in f \text{ and } (a, c_2) \in g \text{ so } (a, c_1) \in f \cup g \\ & \text{so } (a, c_1) \in g \text{ as also } (a, c_2) \in g \text{ so } g \text{ is a function} \\ & \text{so } c_1 = c_2 \end{aligned}$$

$$Q13) R \subseteq A \times B, S \subseteq B \times C$$

$$\text{Dom}(R) = \text{Dom}(S) = B$$

$$\text{so } R: A \rightarrow C$$

$$\begin{aligned} & (b) \quad S: B \rightarrow C \\ & \text{supp } b \in B \\ & \text{Existence: As } b \in \text{Dom}(S) \text{ so } b \in \text{Dom}(S) \text{ so } \exists c \in C \text{ s.t. } (b, c) \in S \\ & \text{Uniqueness: sup } (b, c') \in S \text{ in } b \in B: \text{Range}(R) \text{ so there exists} \\ & \text{one } a \in A \text{ s.t. } (a, b) \in R \text{ as } (a, b) \in R \text{ and } (b, c) \in S \text{ so } (a, c) \in R \text{ so } c = c' \\ & \text{so } (a, c') \in R \text{ but } R \text{ is a function so } c = c' \end{aligned}$$



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S will remain a relation

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- (b)  $R: A \rightarrow B$   $A = \{1, 2\}$ ,  $B = \{1, 2\}$ ,  $C = \{1, 2\}$   
 $R = \{(1, 1), (1, 2), (2, 1)\}$   
 $S = \{(1, 2), (2, 2)\}$   
 $\text{So } R = \{(1, 2), (2, 1)\}$

Q 13) Note: The uniqueness proof for a relation is kind of proof by contradiction in which we contradict the fact that  $S$  is a relation (e.g.  $\beta$ ) so  $(a, b), (a, b_2)$  could also be  $\beta$ . So  $b = b_2$  is must. Also, it's logic of  $\exists!$  at backed that says if 2 elements belong, satisfying some property, they must be equal.

- Q 11)  $\beta: A \rightarrow B$   $S \subseteq B \times B$   
 $R = \{(x, y) \in A \times A \mid (\beta(x), \beta(y)) \in S\}$

(a)  $S$  is reflexive  $\rightarrow R$  is reflexive  
 $\forall b \in B (b, b) \in S \rightarrow \forall a \in A (a, a) \in R$   
 Suppose  $a \in A$  then  $\rightarrow (a, a) \in R$   
 $\exists! b \in B (a, b) \in \beta$  ( $\beta(a) = b$ ) =  $(\beta(a), \beta(a)) \in S$   
~~then~~  $\beta(a) = b \in B$   
 As  $S$  is reflexive,  $(b, b) \in S$  or  $(\beta(a), \beta(a)) \in S$

(b)  $S$  is symmetric  $\rightarrow R$  is symmetric  
 Let  $a_1, a_2 \in R$   $\rightarrow a_1 R a_2$   
 $a_1 R a_2 \rightarrow (\beta(a_1), \beta(a_2)) \in S$

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$\beta$  is symmetric  $\rightarrow \beta(a_1) \in \beta(a_2)$  so by def of  $R$ ,  $a_1 R a_2$ .

(c)  $S$  is transitive  $\rightarrow R$  is transitive  
 Suppose  $a_1, a_2, a_3 \in A$   $\wedge a_1 R a_2 \wedge a_2 R a_3 \rightarrow a_1 R a_3$   
 $\beta(a_1) \in \beta(a_2) \wedge \beta(a_2) \in \beta(a_3)$ . By transitivity of  $S$ ,  $\beta(a_1) \in \beta(a_3)$ .

Q 15)  $\beta: A \rightarrow B$ ,  $R \subseteq A \times A$

$S = \{(x, y) \in B \times B \mid \exists u \in A \wedge (\beta(u) = x \wedge \beta(v) = y \wedge (u, v) \in R)\}$

(a) Already solved in week 7. For  $S$  to be reflexive,  $\forall b \in B (b, b) \in S$ . Note all elements of  $B$  are  $\beta(a)$  for some  $a \in A$ .  
 (c) We will help to see that when  $\beta(a_1) = b_1$  and  $\beta(a_2) = b_2$ , then  $\beta(a_1) = b_1 \in \beta(a_2) = b_2$  but for  $b_2 \in S$ , then we can be  $\beta(a') = b_2$  (not need  $b_2$  again)

Q 16)  $A, B$  are sets.  $R = \{(a, b) \mid a \in A \wedge b \in B\}$   
 $R \subseteq B \times B$ .

$S = \{(b, g) \in B \times B \mid \forall x \in A ((\beta(x), g(x)) \in R)\}$

(a) Yes. Suppose  $\beta \in F \rightarrow (b, b) \in S$ .  
 Let  $x \in A$ . As  $\beta: A \rightarrow B$  so  $\exists! b \in B (\beta(x) = b)$ . As  $R$  is reflexive, so  $(b, b) \in R = (\beta(x), \beta(x)) \in R$  so  $(\beta(x), \beta(x)) \in S$ .

(b) Yes. Suppose  $(\beta(x), \beta(y)) \in S$ . then  $\forall x (\beta(x), g(x)) \in R$



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let  $x \in A \rightarrow (g, f) \in R$

As  $x \in A$ , so  $(f(x), g(x)) \in R$  &  $R$  is symmetric, so  $(g(x), f(x)) \in R$   
so  $(f, g) \in S$ .

(c) Yes. Easy proof. Similar to symmetry.

Q17)  $A \neq \emptyset, f: A \rightarrow A$

$\exists a \in A \forall x \in A (f(x) = a) \rightarrow \forall g: A \rightarrow A (f \circ g = f)$

let  $g: A \rightarrow A$  be arbitrary

$\rightarrow$  Suppose  $(a_1, a_2) \in f \circ g$ . Then  $(f, x) \in g \cap (a_1, a_2) \in f$ . By (1),

$\forall x \in A (f(x) = a_2)$  but also  $f(x) = a_1$  so  $a_1 = a_2$

so  $\forall x \in A (f(x) = a_1)$ . Since  $a_1 \in A$ ,  $(f, a_1) \in f$  so  $(a_1, a_2) \in f$ .

$\leftarrow$  Suppose  $f(a) = a_2$  as  $(a_1, a_2) \in f$ . As  $\forall x \in A (f(x) = a_1)$ , we

know  $a_1 \in A$  so  $f(a_1) = a_2$ . So  $a_1 = a_2$  as  $\forall x \in A (f(x) = a_1)$

Sketch: Do not forget that to prove that 2 functions are

equal, we use the definition  $f = g$  iff  $\forall x [f(x) = g(x)]$

So to prove this we know suppose  $g: A \rightarrow A$

now prove  $\forall x \in A (f \circ g(x) = f(x))$

let  $x \in A$ .  $f \circ g(x) = f(g(x))$  as  $g: A \rightarrow A$

$\exists! a_1 \in A (g(x) = a_1)$  so  $f(g(x)) = f(a_1)$  & as

$f: A \rightarrow A \exists! a \in A (f(a_1) = a)$ . By the given (1),

we can conclude  $\forall g \in A (f(g) = a)$  so as  $x \in A$

$f(x) = a$  as well so  $f \circ g(x) = f(g(x)) = f(a_1) = a = f(x)$

so  $f \circ g(x) = f(x)$

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(b)  $\forall g: A \rightarrow A (f \circ g = f) \rightarrow f$  is constant  
 $\exists a \in A \forall x \in A (f(x) = a)$

$\rightarrow \forall g: A \rightarrow A [\forall x \in A (f \circ g(x) = f(x))]$   
 $\forall g: A \rightarrow A [\forall x \in A (f(g(x)) = f(x))]$

Intuition & understanding

First  $A \neq \emptyset$  so select  $a \in A$ . An important key is that  $f$  is a function, so select  $f(a) = a$ . Last important key is that  $f$  is a function to apply universal instantiation, we have to select any  $g$  that is a function (to use this given). Our only choice is to define  $\forall x \in A (g(x) = a)$  as  $a$  is fully defined here.

Note that this does make  $g$  a constant function but to apply universal instantiation, (which needs for all  $g: A \rightarrow A$ ), we just need one  $g$  that is a function.

Q18)  $K = \{f | f: \mathbb{R} \rightarrow \mathbb{R}\}$

$K = \{ (f, g) \in K \times K | \exists a \in \mathbb{R} \forall x \in \mathbb{R} (f(x) = g(x)) \}$

(a)  $f: \mathbb{R} \rightarrow \mathbb{R} \quad f(x) = |x|$

$g: \mathbb{R} \rightarrow \mathbb{R} \quad g(x) = x$

$(f, g) \in K$

$f \neq g$  are already functions.  $\exists a \in \mathbb{R} \forall x \in \mathbb{R} (f(x) = g(x))$

let  $a = 0$

then  $\forall x \in \mathbb{R} \quad f(x) = |x| = x = g(x)$

(b) Reflexivity & Symmetry

let  $(f, g) \in K$  then  $\exists a \in \mathbb{R} \forall x \in \mathbb{R} (f(x) = g(x))$

which is same as  $\exists a \in \mathbb{R} \forall x \in \mathbb{R} (f(x) = f(x))$  so  $(f, f) \in K$



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Supremacy Reflexivity

$\forall f \in F \{ h \} \{ R \}$

we  $f \in F$

$\exists a \in R \forall x > a (f(x) = h(x))$

Clearly for all  $x$ ,  $f(x) = h(x)$  as  $f$  is a function.

so it must be true for some  $x > a$

Transitivity:

$\forall f, g, h \in F [f R g \wedge g R h \rightarrow f R h]$

Let  $f, g, h \in F$

$\& \text{supp } f R g \wedge g R h \rightarrow f R h$

$f R g$  means  $a_1 \in R \forall x > a_1 (f(x) = g(x))$

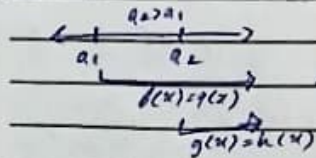
$g R h$  means  $a_2 \in R \forall x > a_2 (g(x) = h(x))$

We have to prove  $\exists a \in R \forall x > a (f(x) = h(x))$

We can find it by looking at number line. If  $a_1 = a_2$  then

$a = a_1 = a_2$  works. If  $a_1 > a_2$  then  $a = a_1$  works.

If  $a_2 > a_1$  then  $a = a_2$  works.



so for  $x > a_2$

both  $f(x) = g(x)$

$\& g(x) = h(x)$  will

hold. similar for other

2 cases

The reason we arrived at cases is that if we look at goal & the givens. We have to prove  $f(x) = h(x)$  & only way that looks is  $f(x) = g(x)$  &  $g(x) = h(x)$  so there should be some  $x$  for which both of them holds true & we can conclude  $f(x) = g(x) = h(x)$ .

That a might be different in case  $a_1 < a_2$  or  $a_1 > a_2$  or  $a_1 = a_2$ .

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(Case 1)  $a_1 = a_2$ . Then let  $a = a_1$ . Clearly  $\forall x > a$

$f(x) = g(x) = h(x)$  so  $f R h$ .

(Case 2)  $a_1 < a_2$ . Then let  $a = a_1$ . Let  $x > a_2$  be arbitrary

As  $x > a_1$  so  $f(x) = g(x)$ . Also  $x > a_2 > a_1$  so  $x > a_1$

so  $g(x) = h(x)$  & so  $f(x) = h(x)$  so  $f R h$

(Case 3)  $a_2 < a_1$ . Then let  $a = a_1$ . Let  $x > a_1$  be arbitrary

As  $x > a_1$  so  $f(x) = g(x)$ . Also  $x > a_1 > a_2$  so  $x > a_2$

so  $g(x) = h(x)$ . Therefore  $f(x) = h(x)$  & thus  $f R h$ .

(Q19) (a) we just need to find one combination of  $a$  &  $\epsilon$  that works. One combination of  $a$  &  $\epsilon$  already implies infinite combinations as example for any  $b > a$ , then some  $\epsilon$  would work.

$F = \{ f \mid f: \mathbb{R}^+ \rightarrow \mathbb{R} \}$

for any  $g \in F$ ,  $O(g) = \{ f \in F \mid \exists a \in \mathbb{R}^+ \exists c \in \mathbb{R}^+ \forall x > a (|f(x)| \leq |g(x)|) \}$

(b)  $f(x) = 7x + 3$ ,  $\mathbb{R}^+ \rightarrow \mathbb{R}$

$g(x) = x^2$ ,  $\mathbb{R}^+ \rightarrow \mathbb{R}$

prove that  $f \in O(g)$

$= \exists a \in \mathbb{R}^+ \exists c \in \mathbb{R}^+ \forall x > a$

Let  $a = 3$  &  $c = 8$

Then  $|f(x)| = |f(x) = 7x + 3| \leq 7x + x = 8x \leq 8x^2 = c|g(x)|$ .

Basically we needed to show  $\forall x > 3 (|7x + 3| \leq 8x^2)$

We can choose any  $a$  &  $\epsilon$  that works but choose one that

is easiest to prove. For a different combination, we

would have to go for proof by contradiction or induction.



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(b) There are 2 ways. Assume  $\exists a \in \mathbb{R}^+ \exists c \in \mathbb{R}^+ \forall x > a (|f(x)| < c|g(x)|)$   
or simply draw the negation that is  
 $\forall a \in \mathbb{R}^+ \forall c \in \mathbb{R}^+ \exists x > a (|g(x)| > c|f(x)|)$

but we do second.

let  $a \in \mathbb{R}^+ \& c \in \mathbb{R}^+$   $\exists x > a (x^2 > c(7x+3))$   
let  $x = 10c$  such that  $10c > a \& 10c > 10c$   $x^2 > c(7x+3)$

since  $10c > 10c$  so  $x^2 > 10cx = x(7x+3x)$   
also  $(7x+3x) > c(7x+3)$

so we find an  $x > a$  that prove the negation of statement

(c)  $f \in O(g)$

$f \in O(g)$

$s, t \in \mathbb{R}$

$f: \mathbb{R}^+ \rightarrow \mathbb{R} \quad f(x) = s f_1(x) + t f_2(x)$

$\rightarrow f \in O(g)$

$= \exists a \in \mathbb{R}^+ \exists c \in \mathbb{R}^+ \forall x > a (|f(x)| \leq c|g(x)|) \quad \text{--- (1)}$

$f_1 \in O(g)$

$= \exists a_1 \in \mathbb{R}^+ \exists c_1 \in \mathbb{R}^+ \forall x > a_1 (|f_1(x)| \leq c_1|g(x)|) \quad \text{--- (2)}$

$f_2 \in O(g)$

$= \exists a_2 \in \mathbb{R}^+ \exists c_2 \in \mathbb{R}^+ \forall x > a_2 (|f_2(x)| \leq c_2|g(x)|) \quad \text{--- (3)}$

To prove the goal, we need to show that  $f(x) = |s f_1(x) + t f_2(x)| \leq c|g(x)|$ . We have to find a  $c$  that satisfies this property for all  $x > a$ . Note we can use  $|f_1(x)| \leq c_1|g(x)|$  if  $x > a_1$  & similarly

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for  $a_2$  so  $x > a_1 \& x > a_2$  should be checked. To check

$c$  note that  $|s f_1(x) + t f_2(x)| \leq |s f_1(x)| + |t f_2(x)|$

Already  $|f_1(x)| \leq c_1|g(x)| \& |f_2(x)| \leq c_2|g(x)|$

so this limits toward  $c_1 + c_2$  as we  $c|g(x)|$  become

$c_1|g(x)| + c_2|g(x)| = |g(x)|$ . Now only problem of  $s$

$\& t$  remains so we can multiply with  $s \& t$  also

but their absolute being they can be negative. so

$c = c_1|s| + c_2|t|$  works. Finally  $s \& t$  can be 0, so

we just add 1 to make sure  $c \in \mathbb{R}^+$ .  $c = c_1|s| + c_2|t| + 1$

For proof is simple now.

let  $a = \max(a_1, a_2)$  ( $\because$  Now  $a > a_1 \& a > a_2$ )

let  $c = c_1|s| + c_2|t| + 1$

let  $x > a$  is arbitrary then also  $x > a_1 \& x > a_2$ . (Now we can use (2) & (3))

then  $|f(x)| = |s f_1(x) + t f_2(x)| \leq |s f_1(x)| + |t f_2(x)|$

$\leq c_1|s||g(x)| + c_2|t||g(x)| = (c_1|s| + c_2|t|)|g(x)|$

$\leq c|g(x)|$ . ( $\because$  as we didn't write +1 so  $\leq$  should be  $<$ )

$\therefore$  we used as  $|f_1(x)| \leq c_1|g(x)|$

Multiply by  $|s| \rightarrow |s||f_1(x)| \leq c_1|s||g(x)|$

(Q20)  $g: A \rightarrow B$

$R = \{ (x, y) \in A \times A \mid g(x) = g(y) \}$

$R$  is equivalence

Reflexivity  $\forall x \in A (x, x) \in R$

Supp  $x \in A$  then clearly  $(x, x) \in A \times A \& g(x) = g(x)$



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so  $(x, y) \in R$ .

Symmetry  $\forall x, y \in A (xRy \rightarrow yRx)$

Supp  $x \in A \wedge y \in A \wedge xRy \rightarrow yRx$

or by means  $g(x) = g(y)$  which is same as  $g(y) = g(x)$

so  $yRx$ .

Transitivity  $\forall x \in A \forall y \in A \forall z \in A [(xRy \wedge yRz) \rightarrow xRz]$

Supp  $x \in A, y \in A, z \in A$

Supp  $xRy \wedge yRz \rightarrow xRz$

which means  $g(x) = g(y) \wedge g(y) = g(z)$  so  $g(x) = g(z)$

so  $xRz$ .

(b)  $R$  is equivalence on  $A$

$g: A \rightarrow A/R \mid g(x) = [x]_R$

Prove that  $R = \{(x, y) \in A \times A \mid g(x) = g(y)\}$

Note that  $R$  is simply the set of pairs  $(x, y)$  where  $[x]_R = [y]_R$

$\forall x \in A \forall y \in A [(x, y) \in R \rightarrow (x, y) \in \{(x, y) \in A \times A \mid g(x) = g(y)\}]$

Supp  $(x, y) \in R$

By symmetry of  $R$ ,  $yRx$   $\oplus$

so  $y \in [x]_R$  by def of  $[x]_R$   $\oplus$

so  $[x]_R = [y]_R$  Lemma 4.5.5.

so  $g(x) = g(y)$  where  $(x, y) \in \{(x, y) \in A \times A \mid g(x) = g(y)\}$

This is a series of equivalent statements that imply both sides.

Note that  $R$  here was like saying  $R = \bigcup_{x \in A/R} x \times x$

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2) Note: The compatibility relation (ex 24, sec 4.5)

tells us that  $R$  is compatible with  $S$  if any 2 elements are related by  $R$  iff their elements of their same equivalence class are also related by  $R$ . So  $R$  respects the transitivity of  $S$ .

The compatibility of functions that any 2 elements who have same element must have same function output.

so  $f$  collapses every class of  $A/R$  to a single output.

(a)  $f: A \rightarrow B$

$f$  is equivalence relation on  $A$

$f$  is compatible with  $R = \forall x \in A \forall y \in A (xRy \rightarrow f(x) = f(y))$

Existence: let  $u = \{(x, y) \in A \times B \mid \exists z \in [x]_R (g(z) = y)\}$

Note: This can be checked by analyzing a small example. Also it's same as saying  $\forall z \in [x]_R (f(z) = y)$

because  $f$  makes sure every element in  $[x]_R$  has same output in  $B$ . But mathematicians use  $\exists$  to prove empty sets & empty proofs.

$f$  is a function:

$\forall [x]_R \in A/R \exists! y \in B (h([x]_R) = y)$

Let  $[x]_R \in A/R$

Existence: let  $y = f(x) \in B$  ( $\because$  use the same  $x$ )

Clearly  $x \in [x]_R$  &  $f(x) = y$   $\oplus$

(Uniqueness of  $y$ : let  $y_1 \in B (h([x]_R) = y_1) \oplus$

let  $y_2 \in B (h([x]_R) = y_2) \oplus$

$\oplus \rightarrow x \in [x]_R (f(x) = y_1)$

$\therefore$  By def of  $h$



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- ②  $\rightarrow x_2 \in [x] \mid f(x_2) = y_2$   
 $\therefore x_1 R x \wedge x_2 R x \quad \therefore$  By def of  $[x]$   
 $\therefore$  By compatibility of  $f$   
 $\text{so } f(x_1) = f(x_2) = f(x)$   
 $\text{so } y_1 = y_2 \quad \therefore f(x_1) = y_1 \wedge f(x_2) = y_2$

Uniqueness of  $h$ :

Supp  $h_1: A/R \rightarrow B$  — ③

$h_2: A/R \rightarrow B$  — ④

$\rightarrow h_1 = h_2$

$$= \forall [x] \in A/R \mid h_1([x]) = h_2([x])$$

Let  $[x] \in A/R \rightarrow h_1([x]) = h_2([x])$

③  $\rightarrow ([x], y_1) \in A/R \times B \mid x_1 \in [x] \mid f(x_1) = y_1$

④  $\rightarrow ([x], y_2) \in A/R \times B \mid x_2 \in [x] \mid f(x_2) = y_2$

$\therefore$  By def of  $h$

$\therefore x_1 R x \wedge x_2 R x \quad \therefore$  By def of  $[x]$

$\therefore$  By compatibility of  $f$

$\therefore f(x_1) = y_1 = y_2 \quad \therefore f(x_1) = y_1 \wedge f(x_2) = y_2$

$\therefore h_1([x]) = y_1 = y_2 = h_2([x])$

Hence Proved!

+++++

⑤ Finally  $\forall x \in A \mid h([x]) = f(x)$

Supp  $x \in A \rightarrow h([x]) = f(x)$

Clearly  $x \in [x]$

As  $h$  is a function from  $A/R \rightarrow B$

$\therefore y \in B \mid \exists x_1 \in [x] \mid f(x_1) = y$

as  $x \in [x]$  so  $f(x) = y = h([x])$

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(b)  $n: A/R \rightarrow B$

$\forall x \in A \mid n([x]) = f(x)$

$\forall x \in A \mid n([x]) = f(x) = f(y)$

Let  $x \in A \wedge y \in A \wedge x R y$

$f(x) = f(y)$

①  $\therefore n([x]) = f(x)$

②  $\therefore n([y]) = f(y)$

But since  $x R y$ , so  $y \in [x]$  so  $[y] = [x]$  — Lemma 4-5.5

so  $n([x]) = f(x) = n([y]) = f(y)$

so  $f(x) = f(y)$

Hence Proved!

22) By making classes & analyzing, we can keep them out to be:

$\{0, 5, 10, 15, \dots\}$

$\{1, 6, 11, 16, 21, 26, \dots\}$

$\{2, 7, 12, 17, 22, 27, \dots\}$

$\{3, 8, 13, 18, 23, 28, \dots\}$

$\{4, 9, 14, 19, 24, 29, \dots\}$

(a) This can be simply proved by defining  $F$  & showing that  $F$  is compatible with  $R$ . The rest of proof will follow from Exercise 21.

The proof that  $F = \{(x, [x]) \in N \times N/R \mid f(x) = [x^2]\}$  is compatible with  $R$  is easy. Supp  $x \in N, y \in N$  &  $x R y$  which means  $[x] = [y]$ . Then show  $[x^2] = [y^2]$  is another proof given on whiteboard. Compatibility will follow from that proof.



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Another comprehensive way is to define  $h$

$$h = \{ ([x], [y]) \in \mathbb{N}/R \times \mathbb{N}/R \mid \exists x_1 \in [x] ([x_1]^2 = [y]) \}$$

This shows  $h$  is a function & the uniqueness of  $h$ .

It can also be defined as (using  $f$ )

$$h = \{ ([x], [y]) \in \mathbb{N}/R \times \mathbb{N}/R \mid \exists x_1 \in [x] (f(x_1) = [y]) \}$$

In both cases, whole proof is similar.

(b) This is a proof by contradiction

Assume  $h$  exists

$$h: \mathbb{N}/R \rightarrow \mathbb{N}/R (\forall x \in \mathbb{N} (h([x]) = [2^x]))$$

$$\text{Then } h([0]) = [2^0] = [1]$$

$$\& h([5]) = [2^5] = [32]$$

$$[0] = [5] \text{ but } [1] \neq [32]. \text{ This contradicts}$$

the fact that  $h$  is a function. ( $[0] = [5]$  so this is same input but we got 2 different outputs  $[1] \neq [32]$ )

Some Notes about  $O(x) \rightarrow Q19$

The big-O is any upper bound (e.g.  $7x + 3 \in O(x)$ ).

Also  $7x + 3 \in O(x^2)$ . In Algorithm analysis, we use  $\Theta$  which shows that it is a tight bound (both upper & lower bound). This shows exact growth rate. Like  $O(x)$  was upper bound,  $\Omega$  was lower bound. i.e

$$f \in \Omega(g) \text{ iff } \exists a \in \mathbb{Z}^+ \exists c \in \mathbb{R}^+ \forall x \geq a (|f(x)| \geq c|g(x)|)$$

$$\text{Now } f \in \Theta(g) \text{ iff } f \in O(g) \wedge f \in \Omega(g)$$

$$\text{iff } f \in O(g) \wedge g \in O(f)$$



